Multimodal Text and Graphics Application

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Since Ollama does not support the text-based image function, we need to use other tools to implement the local text-based image function. Currently, voice control of the text-based image function is not supported, so the content of this article is the same as the offline text multimodal text-based image application.

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1. Concept Introduction

1.1 What is Text-to-Image?

Text-to-Image is an AI technology that automatically generates corresponding images based on **text description**. Simply input a piece of text (for example, "A Shiba Inu wearing sunglasses surfing on the beach"), and the AI model will generate an image that matches the description based on semantic understanding, without any drawing or design knowledge required.

Core Principles

1. **Language Understanding**: The AI first parses your text and identifies keywords (such as "Shiba Inu," "sunglasses," and "beach"). 2. Image Generation: Based on training from massive amounts of image data, AI converts text into visual elements and combines them into a coherent image.
2. Style Adaptation: You can specify a style (realistic, cartoon, watercolor, etc.), and the AI will adjust the generated image quality.

1.2 What is FastSDCPU?

FastSDCPU is an open-source Stable Diffusion image processing project optimized for CPU devices. Through algorithmic and engineering optimizations, it enables rapid generation of high-quality images on standard computers without GPUs, significantly lowering the hardware barrier to entry for AI painting.

Core Features

1. Pure CPU Operation: No dedicated graphics card required, allowing it to run directly on laptops, desktops, or edge devices.
2. High-Speed Inference: Incorporating technologies such as Latent Consistency Models (LCM), image generation takes only 4–8 steps, significantly reducing latency. 3. **Lightweight Deployment**: Supports model quantization (such as INT8) and OpenVINO acceleration, reducing resource usage and improving response efficiency.

Applicable Scenarios

- Localized AI Creation Tool
- Intranet Image Generation Service
- Teaching Demonstration and Prototyping
- Lightweight Deployment in Resource-Constrained Environments

2. Project Deployment

2.1 Deployment Environment

Open a terminal and execute the following code:

```
# If Git is not installed on your system, run:
sudo apt update
sudo apt install git -y
sudo apt install python3.10-venv -y

# Add environment variables
echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc

# Clone the project code
git clone https://github.com/rupeshs/fastsdcpu.git
cd fastsdcpu

# Create a virtual environment and install dependencies
python -m venv venv
source venv/bin/activate

# Install uv
curl -Ls https://astral.sh/uv/install.sh | sh (This step may fail if you don't
have a proxy in China. If not, skip this step and proceed to the next command.)

#If the curl command in the previous step failed to install uv, run these three
commands:
wget https://mirrors.huaweicloud.com/astral/uv/0.8.4/uv-aarch64-unknown-linux-gnu
-O ~/.local/bin/uv
chmod +x ~/.local/bin/uv
~/.local/bin/uv --version

chmod +x install.sh start-webui.sh
./install.sh --disable-gui
```

Installation successful. Press any key to exit:

```
 + tomesd==0.1.3
 + tomli==2.2.1
 + tomlkit==0.12.0
 - torch==2.7.1+cpu
 + torch==2.5.1
 + torchvision==0.20.1
 + tqdm==4.67.1
 + transformers==4.48.0
 + typer==0.16.0
 - typing-extensions==4.12.2
 + typing-extensions==4.8.0
 + tzdata==2025.2
 + urllib3==2.5.0
 + uvicorn==0.35.0
 + websockets==12.0
 + xxhash==3.5.0
 + yarl==1.20.1
 + zipp==3.23.0
FastSD CPU installation completed,press any key to continue...
```

2.2 LAN Access

Before starting, you need to modify a file to support LAN access. Otherwise, the webui can only be accessed locally:

```
vim ~/fastsdcpu/src/frontend/webui/ui.py
```

After opening the ui.py file, scroll to the last line and find **webui.launch(share=share)**. Change it to **webui.launch(server_name="0.0.0.0",share=share)**

Save the file.

Start:

```
cd ~/fastsdcpu
source venv/bin/activate
./start-webui.sh
```

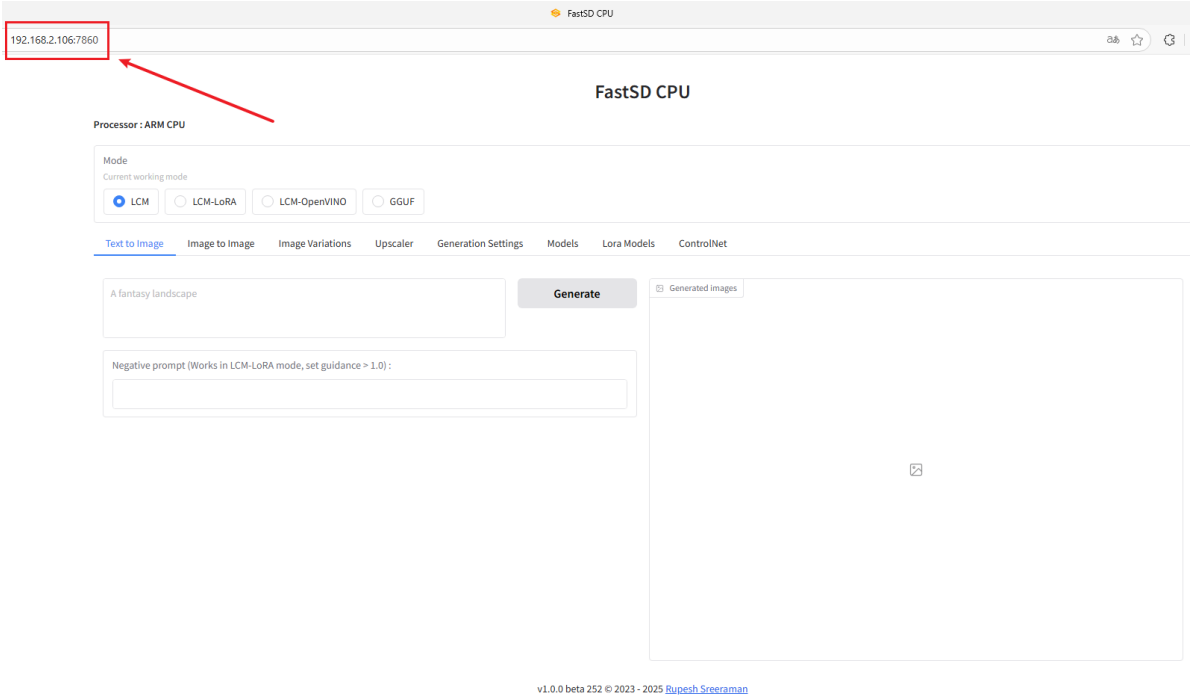
```
Found 7 stable diffusion models in config/stable-diffusion-models.txt
Found 4 LCM-LoRA models in config/lcm-lora-models.txt
Found 10 OpenVINO LCM models in config/openvino-lcm-models.txt
/home/sunrise/fastsdcpu/env/lib/python3.11/site-packages/controlnet_aux/segment_anything/modeling/tiny_vit_sam.py:654:
 UserWarning: Overwriting tiny_vit_5m_224 in registry with controlnet_aux.segment_anything.modeling.tiny_vit_sam.tiny_
vit_5m_224. This is because the name being registered conflicts with an existing name. Please check if this is not exp
ected.
  return register_model(fn_wrapper)
/home/sunrise/fastsdcpu/env/lib/python3.11/site-packages/controlnet_aux/segment_anything/modeling/tiny_vit_sam.py:654:
 UserWarning: Overwriting tiny_vit_11m_224 in registry with controlnet_aux.segment_anything.modeling.tiny_vit_sam.tiny
_vit_11m_224. This is because the name being registered conflicts with an existing name. Please check if this is not e
xpected.
  return register_model(fn_wrapper)
/home/sunrise/fastsdcpu/env/lib/python3.11/site-packages/controlnet_aux/segment_anything/modeling/tiny_vit_sam.py:654:
 UserWarning: Overwriting tiny_vit_21m_224 in registry with controlnet_aux.segment_anything.modeling.tiny_vit_sam.tiny
_vit_21m_224. This is because the name being registered conflicts with an existing name. Please check if this is not e
xpected.
  return register_model(fn_wrapper)
/home/sunrise/fastsdcpu/env/lib/python3.11/site-packages/controlnet_aux/segment_anything/modeling/tiny_vit_sam.py:654:
 UserWarning: Overwriting tiny_vit_21m_384 in registry with controlnet_aux.segment_anything.modeling.tiny_vit_sam.tiny
_vit_21m_384. This is because the name being registered conflicts with an existing name. Please check if this is not e
xpected.
  return register_model(fn_wrapper)
/home/sunrise/fastsdcpu/env/lib/python3.11/site-packages/controlnet_aux/segment_anything/modeling/tiny_vit_sam.py:654:
 UserWarning: Overwriting tiny_vit_21m_512 in registry with controlnet_aux.segment_anything.modeling.tiny_vit_sam.tiny
_vit_21m_512. This is because the name being registered conflicts with an existing name. Please check if this is not e
xpected.
  return register_model(fn_wrapper)
Starting web UI mode
No lora models found, please add lora models to /home/sunrise/fastsdcpu/lora_models directory
/home/sunrise/fastsdcpu/env/lib/python3.11/site-packages/gradio/components/dropdown.py:226: UserWarning: The value pas
sed into gr.Dropdown() is not in the list of choices. Please update the list of choices to include:  or set allow_cust
om_value=True.
  warnings.warn(
/home/sunrise/fastsdcpu/env/lib/python3.11/site-packages/gradio/components/base.py:194: UserWarning: show_label has no
 effect when container is False.
  warnings.warn("show_label has no effect when container is False.")
* Running on local URL:  http://0.0.0.0:7860

To create a public link, set `share=True` in `launch()`.
```

You can then access the webui by entering your motherboard's IP address: 7860 in your browser.

2.3 Using the Vincent Image Function

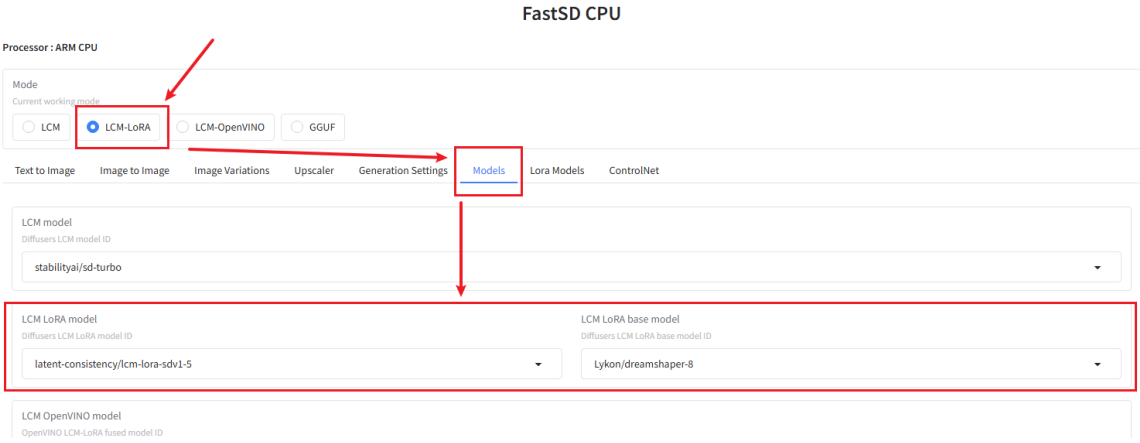
Use ifconfig in the terminal to query the motherboard's IP address. For example, mine is 192.168.2.106.

Then open a browser and enter **your motherboard's IP address:7860** . For example, I entered 192.168.2.106:7860. This will access the web UI.

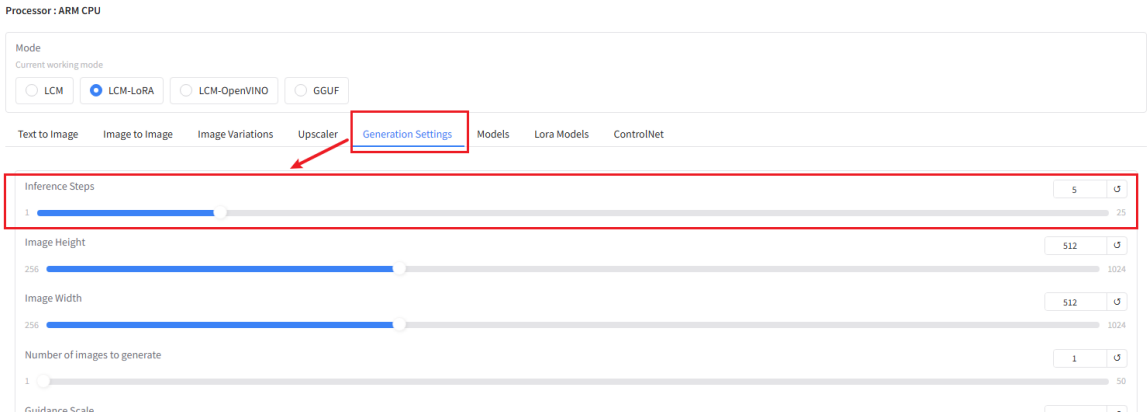


Next, click LCM-LoRA. This model uses less memory. If you want to use a different model, feel free to research it yourself.

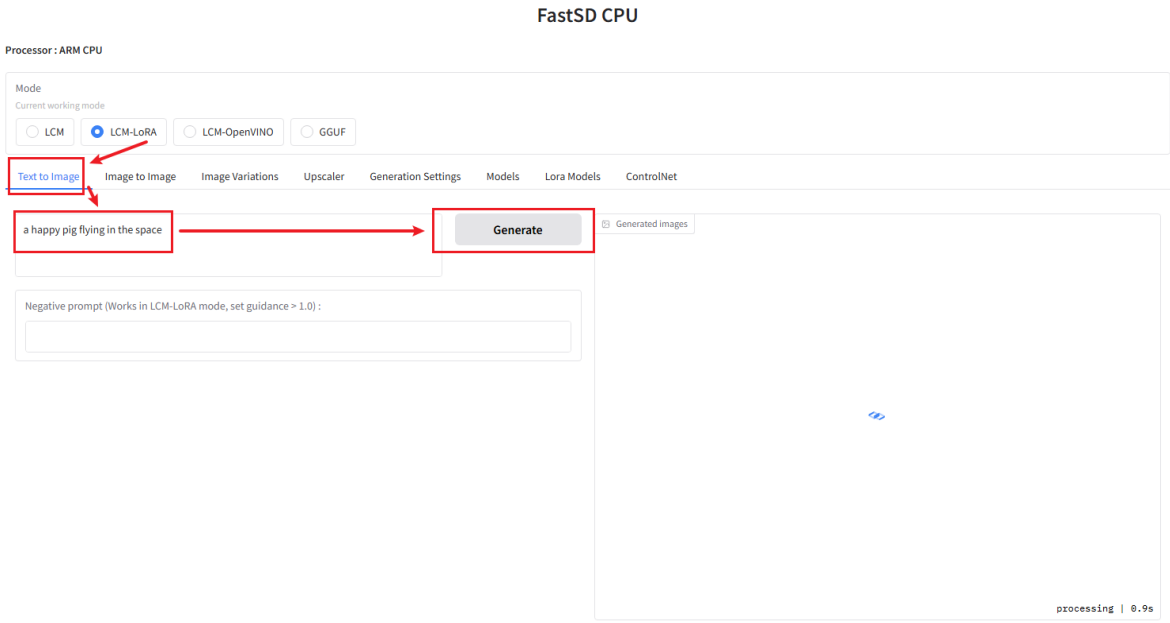
Next, click Models to view the LCM-LoRA model settings. You can change the model to your preference, or just stick with the default, as I did.



Next, click Generation Settings and increase the Inference Steps setting to improve the quality of the generated image. I've set it to 5.



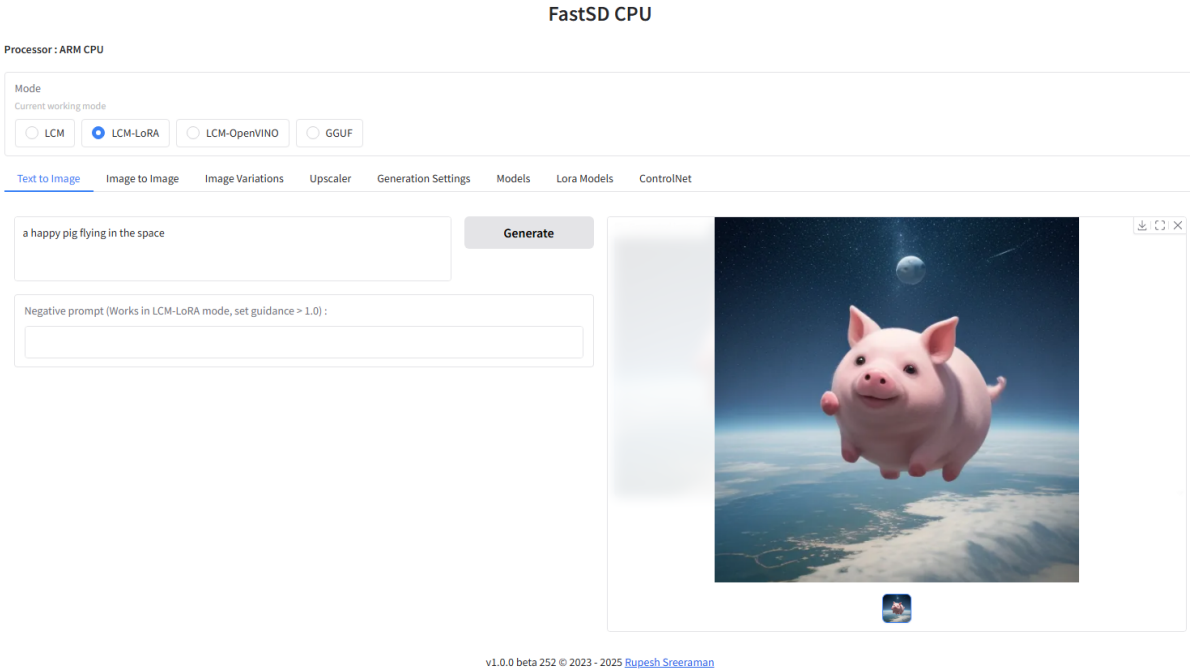
Next, return to the Text to Image dialog box and enter the content you want to generate. Click Generate to begin generating the image.



For first-time users, you'll need to download the model. You'll see the default model being downloaded in the terminal. Once it's finished downloading, the text-to-image function will begin.

```
'init_image': None,
'lcm_lora': {'base_model_id': 'Lykon/dreamshaper-8',
             'lcm_lora_id': 'latent-consistency/lcm-lora-sdv1-5'},
'lcm_model_id': 'stabilityai/sd-turbo',
'lora': {'enabled': False,
         'fuse': True,
         'models_dir': '/home/sunrise/fastsdcpu/lora_models',
         'path': '',
         'weight': 0.5},
'negative_prompt': '',
'number_of_images': 1,
'openvino_lcm_model_id': 'rupeshs/sd-turbo-openvino',
'prompt': 'a happy pig flying in the space',
'rebuild_pipeline': False,
'seed': 123123,
'strength': 0.6,
'token_merging': 0.0,
'use_gguf_model': False,
'use_lcm_lora': True,
'use_offline_model': False,
'use_openvino': False,
'use_safety_checker': False,
'use_seed': False,
'use_tiny_auto_encoder': False}
***** Init LCM-LoRA pipeline - Lykon/dreamshaper-8 *****
model_index.json: 100%|| 642/642 [00:00<00:00, 2.80MB/s]
config.json: 100%|| 796/796 [00:00<00:00, 3.81MB/s]
config.json: 100%|| 724/724 [00:00<00:00, 3.74MB/s]
special_tokens_map.json: 100%|| 472/472 [00:00<00:00, 2.27MB/s]
merges.txt: 525kB [00:00, 6.20MB/s]| 0.00/472 [00:00<?, ?B/s]
preprocessor_config.json: 100%|| 520/520 [00:00<00:00, 2.53MB/s]
tokenizer_config.json: 100%|| 737/737 [00:00<00:00, 3.54MB/s]
config.json: 1.87kB [00:00, 5.15MB/s]| 0.00/1.22G [00:00<?, ?B/s]
vocab.json: 1.06MB [00:00, 9.26MB/s]| 0.00/520 [00:00<?, ?B/s]
scheduler_config.json: 100%|| 614/614 [00:00<00:00, 2.99MB/s]
config.json: 100%|| 756/756 [00:00<00:00, 3.99MB/s]
model.safetensors:   2%|| 10.5M/492M [00:00<00:23, 20.3MB/s]
model.safetensors:  32%|| 157M/492M [00:36<01:24, 3.98MB/s]
model.safetensors:  12%|| 147M/1.22G [00:34<04:28, 3.98MB/s]
diffusion_pytorch_model.safetensors:  41%|| 136M/335M [00:34<00:50, 3.96MB/s]
diffusion_pytorch_model.safetensors:   4%|| 136M/3.44G [00:34<13:52, 3.97MB/s]
```

Generated result:



This project has improved support for English, and the generated images will be more consistent with the text. We recommend using English descriptions when generating images.